

Major Project Report

On

Purchase Order System

(A dissertation submitted in fulfillment of the requirements of 6th semester,
Bachelor of Computer Applications (BCA) of
Maulana Abul Kalam Azad University of Technology, West Bengal)

Submitted by

Name	University Roll No.
Ushnish Tapaswi	23401222013
Obada Saquib	23401222070

Under the guidance of
Dr. Suparna Dhar
Department of Computing & Analytics



**NSHM College of Management & Technology,
Kolkata (234)**

(Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal)
2021-2022

CERTIFICATE

This is to certify that the project report is on
“Purchase Order System – Web Application with Admin Portal”

a record of Bonafide work, carried out by

Mr. Ushnish Tapaswi
Mr. Obada Saquib

Co-author
Mr. Ushnish Tapaswi

Co-author
Mr. Obada Saquib

Signature of Supervisor
Dr. Suparna Dhar

Signature of Head, Dept. of Computing & Analytics
NSHM College of Management & Technology, Kolkata
Dr. Suparna Dhar

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to everyone who contributed to the successful completion of the Purchase Order System and its Software Requirement Specification Report. This project represents a collaborative effort, and its development would not have been possible without the guidance and support of several individuals.

First and foremost, we extend our sincere thanks to **Dr. Suparna Dhar**, NSHM Knowledge Campus - Kolkata, for their invaluable guidance, constructive feedback, and encouragement throughout the project. Their expertise and insights greatly enhanced the clarity and quality of this project and its document.

We are also grateful to **NSHM Knowledge Campus – Kolkata** for providing us with the resources and a conducive environment to carry out this project effectively.

Additionally, we would like to acknowledge our peers and colleagues for their helpful discussions, suggestions, and moral support during the development of this project and drafting this document.

Finally, we appreciate the opportunity to work on this project, which has allowed us to enhance our skills, gain practical knowledge, and contribute meaningfully to a real-world application.

Thank you to everyone who played a role in the completion of this document.

Name	University Roll No.	Signature
Ushnish Tapaswi	23401222013	
Obada Saquib	23401222	

Table of Contents

1. INTRODUCTION	<i>Page 6</i>
Purpose	
Scope	
Overview	
2. GENERAL DESCRIPTION	<i>Page 9</i>
Product Perspective	
Product Functions	
User Characteristics	
Assumptions and Dependencies	
3. SPECIFIC REQUIREMENTS	<i>Page 13</i>
EXTERNAL INTERFACE REQUIREMENTS	
User Interfaces	
Hardware Interfaces	
Software Interfaces	
Communications Interfaces	
FUNCTIONAL REQUIREMENTS	
User Authentication and Authorization	
Purchase Requisition Management	
Vendor Management	
Request for Quotes (RFQ)	
Purchase Order Creation	
Approval Workflow	
Notifications and Alerts	
Reporting and Analytics	
USE CASES	

General User – Create Purchase Requisitions
General User – Track Purchase Requisition Status
Admin User – Manage Purchase Requisitions
Admin User – Generate Purchase Order
Admin User – View Analytics
Admin User – Vendor Management

CLASSES / ATTRIBUTES

User
Vendor
PurchaseRequisition
PurchaseOrder

NON-FUNCTIONAL REQUIREMENTS

Performance
Reliability
Availability
Security
Portability

INVERSE REQUIREMENTS

DESIGN CONSTRAINTS

LOGICAL DATABASE REQUIREMENTS

OTHER REQUIREMENTS

4. ANALYSIS MODELS

Page 34

Sequence Diagram

Er Diagram

5. PREVIEW

Page 39

6. APPENDIX

Page 49

Chapter 1: *Introduction*

1. Introduction

This Software Requirements Specification (SRS) document outlines the requirements for the development and implementation of a **Purchase Order System (POS)** for *NSHM Knowledge Campus, Kolkata*. The system aims to enhance the efficiency and transparency of the procurement process by *systematically record keeping* the creation, *and approval of* purchase orders. By integrating with existing financial systems, the POS will streamline procurement activities *and* ensure compliance with budgetary constraints.

The Purchase Order System will support various user roles - **General Users, and Administrators**. It will facilitate the entire procurement process, from the initiation of purchase requisitions to the final approval and *fulfilment* of purchase orders.

1.1 Purpose

The purpose of this document is to provide a comprehensive Software Requirements Specification (SRS) for the Purchase Order System (POS) being developed for *NSHM Knowledge Campus, Kolkata*. This software aims to *systematically manage and record keep* the procurement process, ensuring efficient and transparent management of purchase orders.

This document outlines the functional and non-functional requirements of the system, providing a clear and detailed specification for developers and stakeholders to ensure a successful implementation.

1.2 Scope

The scope of the Purchase Order System includes the following components.

1.2.1 User Management

Implementing user authentication and authorization mechanisms to secure access to the system.

1.2.2 Purchase Requisition Management

Allowing users to create, edit, submit, and track purchase requisitions.

1.2.3 Vendor Management

Managing vendor information and facilitating vendor selection and communication.

1.2.4 Purchase Order Creation

Automating the generation of purchase orders based on approved requisitions and negotiated terms.

1.2.5 Approval Workflow

Implementing a multi-level approval process for purchase requisitions and orders.

1.2.6 Reporting and Analytics

Providing tools for generating reports on procurement activities and spending.

1.3 Overview

This document is structured as follows.

Section 2 provides a general description of the system, **Section 3** details specific requirements, including functional and non-functional requirements, and **Section 4** presents analysis models. **Section 5** outlines the change management process, and the appendices provide additional supporting information.

The Purchase Order System aims to provide an efficient solution for managing *and record keeping* the procurement process. *The* POS will enhance transparency, accountability, and efficiency in purchase order management. This document details the system's requirements, including functional and nonfunctional aspects.

Chapter 2: *General Description*

2. General Description

This section of the SRS describes the general factors that affect the product and its requirements. This section does not state specific requirements; it only makes those requirements easier to understand.

2.1 Product Perspective

The Purchase Order System (POS) is a **web-based application** designed to efficiently manage and record keep the procurement process at NSHM Knowledge Campus, Kolkata. The system may initiate emails for vendor communication and price negotiation and stores data in a centralized database, which may include, Purchase requisition details, approval details, vendor details, and Purchase Order details.

2.2 Product Functions

The primary functions of the unmanned weighbridge software include:

2.2.1 Purchase Requisition Management

Allow users to create, submit, and track purchase requisitions for goods and services.

2.2.2 Vendor Management

Store and manage vendor information, including contact details.

2.2.3 Request for Quotes (RFQ)

Enable the procurement team to *initiate emails to* request and compare quotes from multiple suppliers.

2.2.4 Purchase Order Creation

Automatically generate purchase orders based on approved requisitions and negotiated terms with vendors.

2.2.5 Approval Workflow

Implement a multi-level approval process for requisitions and purchase orders, ensuring compliance with organizational policies.

2.2.8 Reporting and Analytics

Provide tools for generating detailed reports on procurement activities and spending.

2.3 User Characteristics

This subsection of the SRS describes the general characteristics of the eventual users of the product that will affect the specific requirements. The primary users of the system include:

2.3.1 General User

Responsible for creating Purchase Requisitions, including details of goods/service which may include the following data – Product Type, Quantity, Estimated Costs, and Potential Vendors/Suppliers.

2.3.2 Administrator

Responsible for managing the system, including Purchase Management and Generation of POs, view analytics, and overseeing the procurement process. They need to comply with Organizational policies and fulfil the following roles.

Standalone functions aside, the Administrator may exercise functions of a General User as well, e.g., creation of Purchase Requisitions.

Admin's Role(s)

2.3.2.1 Purchase Management

Manage purchase requisitions and orders (approve/reject), negotiate with vendors, select vendors, and ensure compliance with budgetary constraints. They need to be familiar with procurement processes.

2.3.2.2 Generation of PO

Generation of Purchase Order after approving purchase requisition, selecting vendor, and retrieving pricing details including Unit Price, discounts and tax rates.

2.4 Assumptions and Dependencies

2.4.1 Database Server

A functional and accessible database server is necessary for storing and retrieving data.

2.4.2 User Training

Users are assumed to have received adequate training to use the system effectively.

2.4.3 Web Hosting

The web interface relies on a stable hosting environment to provide real-time monitoring and reporting.

2.4.4 Stable Network Connection

A stable network connection is essential for accurate data capture and system functionality.

Chapter 3: *Specific Requirements*

3. Specific Requirements

This section provides detailed descriptions of the functional and non-functional requirements for the Purchase Order System (POS). It includes the system's external interface requirements, functional requirements, and performance requirements to ensure a comprehensive understanding of the system's capabilities and constraints.

3.1 External Interface Requirements

3.1.1 User Interfaces

The Purchase Order System will include a web-based user interface that provides various functionalities for different user roles.

The components of the user interface *may include the following*.

3.1.1.1 Login Screen

Allows users to log in using *their credentials*.

3.1.1.2 Register Screen

Allows users to send request for account creation.

3.1.1.3 Dashboard

Displays a summary of *product requisitions including pending and approved ones with vendor details and options to generate POs. Displays Approved vendor list along with product catalogue.*

Provides quick access to key functionalities based on the user's role.

3.1.1.4 Purchase Requisition Form

Enables users to create and submit purchase requisitions.

Includes fields for item details, quantity, justification, *and other details as may be necessary including estimated costs or potential vendors.*

3.1.1.5 Vendor Management Screen

Displays a list of approved vendors with their contact details and product *catalogues*.

Allows authorized users to add, edit, or remove vendors.

3.1.1.6 RFQ Management Screen

Enables the procurement team to *initiate emails for* Requests for Quotes (RFQs).

3.1.1.7 Approval Workflow Screen

Provides approvers with a list of requisitions and orders awaiting their review.

Allows approvers to approve, reject, or request modifications to submissions.

3.1.1.8 Reporting and Analytics Dashboard

Offers tools for generating reports on procurement activities *and* spending

Includes options for exporting reports *or POs* in various formats (e.g., PDF, Excel).

3.1.2 Hardware Interfaces

The Purchase Order System will interface with various hardware components to ensure seamless operation and data exchange. *The hardware interfaces may include the following components.*

3.1.2.1 Servers

The database server must be robust and reliable to ensure data integrity and availability.

3.1.2.2 User Devices

The system will be accessible from standard web browsers on desktop computers, laptops, and mobile devices.

Devices must have a stable internet connection to access the system.

3.1.3 Software Interfaces

3.1.3.1 Vendor Communication:

The POS *may contain options to initiate emails* with vendors for RFQ submissions, quote comparisons, and order confirmations.

3.1.3.2 Email and Notification Service:

The POS *may be* integrated with an email and notification service to send alerts and updates to users.

The integration will ensure timely communication of approvals, rejections, and order status changes.

3.1.4 Communications Interfaces

The POS will require various communication interfaces to ensure smooth data exchange and user interactions. Key communication interfaces include:

3.1.4.1 HTTP/HTTPS

The primary protocol for communication between the webbased user interface and the application server.

Ensures secure data transmission using HTTPS.

3.1.4.2 RESTful APIs

Used for integrating with external systems *if required*.

3.1.4.3 Mail

Used for sending email notifications to users regarding approvals, rejections, and order status changes.

Ensures reliable delivery of critical updates.

3.2 Functional Requirements

The functional requirements define the specific behaviors and functionalities that the Purchase Order System (POS) must exhibit. These requirements detail the interactions between the system and its users, as well as the processes the system must support.

3.2.1 User Authentication and Authorization

- ***User Registration***

The System must allow users to send request for account creation after validating phone number or email.

The System must allow authorized users/administrator to approve account creation requests, following which, user may be redirected to an account creation page.

- **User Login**

The system must allow users to log in using a unique username and password.

The system must validate the credentials against stored user data.

Users must be able to reset their password through a secure process.

- ***User Roles and Permissions***

The system must support different user roles, including Admin, Purchase Manager, Approver, and Supplier.

Each role must have specific permissions for accessing and performing certain functions within the system.

- ***User Management***

Admin users must be able to create, update, and delete user accounts.

Admin users must be able to assign roles and permissions to user accounts.

3.2.2 Purchase Requisition Management

- ***Create Purchase Requisition***

Users must be able to create a new purchase requisition by filling out a form with required details, including item description, quantity, budget code, and justification.

- ***Submit Purchase Requisition***

Users must be able to submit the completed requisition form for approval.

The system must notify the relevant approvers of the new requisition submission.

- ***Track Purchase Requisition Status:***

Users must be able to view the status of their submitted requisitions (e.g., pending approval, approved, rejected).

The system must provide updates on any changes in the requisition status.

3.2.3 Vendor Management

- ***Add and Update Vendor Information***

Authorized users must be able to add new vendors to the system, including contact details and product catalogs.

Users must be able to update existing vendor information.

- ***View Vendor List***

Users must be able to view a list of approved vendors, including their contact details and available products.

- ***Deactivate Vendor***

Authorized users must be able to deactivate a vendor, making them unavailable for new purchase orders but retaining their information for historical reference.

3.2.4 Request for Quotes (RFQ)

- ***Create and Send RFQ:***

Purchase Managers must be able to *initiate emails for* a RFQ, specifying the items and quantities required.

The system must send the RFQ to selected vendors.

- ***Select Vendor and Finalize Terms***

Purchase Managers must be able to select a vendor based on the received quotes and finalize the terms of the purchase.

3.2.5 Purchase Order Creation

- ***Generate Purchase Order***

The system must automatically generate a purchase order based on approved requisitions and finalized terms with the selected vendor.

The purchase order must include all relevant details such as item description, quantity, price, delivery terms, and payment terms.

- ***Send Purchase Order to Vendor***

The system *may* send the purchase order to the selected vendor and notify relevant *users of the software*.

3.2.6 Approval Workflow

- ***Multi-Level Approval Process***

The system must support a multi-level approval process for purchase requisitions and orders.

The system must route requisitions and orders to the appropriate approvers based on predefined rules.

- ***Approve or Reject Requests***

Approvers must be able to review, approve, or reject purchase requisitions and orders.

Approvers must be able to request modifications or additional information before making a decision.

- ***Notify Stakeholders***

The system must notify relevant stakeholders of approval decisions and any required actions.

3.2.7 Notifications and Alerts

- ***System Notifications***

The system must send notifications to users for various events, such as requisition submissions, approvals, rejections, and order status changes.

- ***Email Alerts***

The system must send email alerts to users for critical updates and actions required.

Users must be able to customize their notification preferences.

3.2.8 Reporting and Analytics

- ***Generate Reports***

The system must provide tools for generating detailed reports on procurement activities, *and spending*.

Reports must be customizable based on user-defined parameters.

- ***Export Reports***

Users must be able to export reports in various formats, including PDF and Excel.

The system must ensure the accuracy and completeness of exported data.

3.3 Use Cases

3.3.1 General User

3.3.1.1. Use Case 1: Create Purchase Requisition

Actor: General User

Preconditions: User is logged into the system.

Postconditions: Purchase requisition is submitted for approval.

Main Success Scenario:

1. The user navigates to the "Create Purchase Requisition" form.
2. The user enters the product type, quantity, estimated costs, and potential vendors/suppliers.
3. The user attaches any supporting documents.
4. The user submits the purchase requisition.
5. The system validates the entered information and saves the requisition.
6. The system notifies the relevant approvers of the new requisition.

Extensions:

- The user inputs incomplete or invalid data:

The system prompts the user to correct the errors.

- The system encounters an error while saving the requisition:

The system displays an error message and prompts the user to retry.

3.3.1.2 Use Case 2: Track Purchase Requisition Status

Actor: General User

Preconditions: User is logged into the system and has submitted one or more requisitions.

Postconditions: User views the status of submitted requisitions.

Main Success Scenario:

1. The user navigates to the "My Requisitions" page.
2. The system displays a list of the user's submitted requisitions with their current status (e.g., pending approval, approved, rejected).
3. The user selects a requisition to view detailed status and history.
4. The system displays the detailed status and any comments from approvers.

Extensions:

- The system encounters an error while retrieving the requisitions:

The system displays an error message and prompts the user to retry.

3.3.2 Administrator

3.3.2.1 Use Case 3: Manage Purchase Requisitions

Actor: Administrator

Preconditions: Administrator is logged into the system.

Postconditions: Requisition is approved, rejected, or sent back for modifications.

Main Success Scenario:

1. The administrator navigates to the "Manage Requisitions" page.
2. The system displays a list of pending requisitions.
3. The administrator selects a requisition to review.
4. The system displays the details of the selected requisition.
5. The administrator reviews the requisition and decides to approve, reject, or request modifications.
6. The administrator submits their decision.

7. The system updates the status of the requisition and notifies the relevant users.

Extensions:

- The administrator encounters an error while reviewing the requisition:

The system displays an error message and prompts the administrator to retry.

- The administrator decides to request modifications:

The system allows the administrator to add comments and sends the requisition back to the original submitter.

3.3.2.2 Use Case 4: Generate Purchase Order

Actor: Administrator

Preconditions: Requisition is approved, and vendor selection is finalized.

Postconditions: Purchase order is generated and sent to the selected vendor.

Main Success Scenario:

1. The administrator navigates to the "Generate Purchase Order" page.
2. The system displays a list of approved requisitions ready for PO generation.
3. The administrator selects an approved requisition.
4. The system retrieves pricing details, including unit price, discounts, and tax rates from the vendor.
5. The administrator reviews the PO details and makes any necessary adjustments.
6. The administrator confirms and generates the PO.
7. The system saves the PO and sends it to the selected vendor.
8. The system updates the requisition status and notifies the relevant

users. **Extensions:**

- The system encounters an error while retrieving pricing details:

The system displays an error message and prompts the administrator to retry.

- The administrator encounters an error while generating the PO:

The system displays an error message and prompts the administrator to retry.

3.3.2.3 Use Case 5: View Analytics

Actor: Administrator

Preconditions: Administrator is logged into the system.

Postconditions: Administrator views procurement analytics.

Main Success Scenario:

1. The administrator navigates to the "Analytics" dashboard.
2. The system displays various metrics and reports on procurement activities, spending, and vendor performance.
3. The administrator selects specific reports or metrics to view detailed information.
4. The system displays the selected detailed report or metric.

Extensions:

- The system encounters an error while retrieving analytics data:

The system displays an error message and prompts the administrator to retry.

3.3.2.4 Use Case 6: Vendor Management

Actor: Administrator

Preconditions: Administrator is logged into the system.

Postconditions: Vendor information is added, updated, or deactivated.

Main Success Scenario:

1. The administrator navigates to the "Vendor Management" page.
2. The system displays a list of current vendors.

3. The administrator selects an option to add a new vendor, update existing vendor details, or deactivate a vendor.
4. The system presents the appropriate form or interface for the selected action.
5. The administrator completes the form or makes the necessary changes.
6. The administrator submits the form or changes.
7. The system updates the vendor information accordingly and confirms the update.

Extensions:

- The administrator encounters an error while adding, updating, or deactivating a vendor:

The system displays an error message and prompts the administrator to retry.

3.4 Classes / Attributes

In this section, classes and their attributes are defined to represent entities within the system. These classes do not include methods but are used to specify the schema of the database.

3.4.1 Class: User

Purpose: To store information about the users of the system, including general users and administrators.

Attributes:

- `user_id` (String): Unique identifier for the user.
- `username` (String): User's login name.
- `password` (String): Encrypted password for user authentication.
- `role` (String): Role of the user (e.g., "General User", "Administrator").
- `name` (String): Full name of the user.
- `username` (String): username of the user.
- `email` (String): Email address of the user
- `password`

3.4.2 Class: Vendor

Purpose: To store information about vendors supplying goods and services.

Attributes:

- `vendor_id` (String): Unique identifier for the vendor. ◦ `name` (String): Name of the vendor.
- `Vendor_email_id` (String): Email address of the contact person.
 - `Vendor_contact_person` (String): contact person of the vendor.
- `Vendor_contact` (String): contact of the vendor. ◦ `Vendor_address` (String): Address of the vendor. ◦ `Vendor_GSTIN` (String): GSTIN of the vendor. ◦ `Vendor_VAT` (String): VAT No. of the vendor. ◦ `Vendor_TIN` (String): TIN No. of the vendor.

3.4.3 Class: PurchaseRequisition

Purpose: To store details of purchase requisitions created by users.

Attributes:

- `requisition_id` (String): Unique identifier for the purchase requisition. ◦ `user_id` (String): Identifier of the user who created the requisition. ◦ `items` (Array of Objects): List of items requested, each containing: `product_type` (String): Type of product or service.
- `quantity` (Number): Quantity requested. ◦ `estimated_cost` (Number): Estimated cost of the item. ◦ `potential_vendor` (String): Potential vendor IDs.
- `status` (String): Current status of the requisition (e.g., "Pending", "Approved", "Rejected").

Purchase Order System

- `comments` (Array of Objects): List of comments, each containing:
 - `user_id` (String): Identifier of the user who made the comment.
 - `comment` (String): The comment text.
- `timestamp` (Date): When the comment was made.
- `created_at` (Date): Timestamp when the requisition was created.
- `required_on` & `required_at` (Date): Timestamp when the requisition is required.

3.4.4 Class: **PurchaseOrder**

Purpose: To store details of purchase orders generated from approved requisitions.

Attributes:

- `order_id` (String): Unique identifier for the purchase order.
- `requisition_id` (String): Identifier of the associated purchase requisition.
- `vendor_id` (String): Identifier of the vendor selected for the order.
 - `items` (Array of Objects): List of items ordered, each containing:
 - `product_type` (String): Type of product or service.
 - `quantity` (Number): Quantity ordered.
 - `unit_price` (Number): Unit price of the item.
 - `discount` (Number): Discount applied to the item.
 - `tax_rate` (Number): Tax rate applicable to the item.
 - `total_cost` (Number): Total cost of the purchase order.
- `status` (String): Current status of the order (e.g., "Pending", "Sent", "Completed").
 - `created_at` (Date): Timestamp when the order was created.
 - `updated_at` (Date): Timestamp when the order was last updated.

3.5 Non-Functional Requirements

3.5.1 Performance

Requirement: The system should be able to handle multiple users without performance degradation.

Measurement Criteria: Response time for any operation (e.g., creating a purchase requisition, generating a purchase order) should not exceed 2 seconds under normal load conditions.

3.5.2 Reliability

Requirement: The system must provide reliable operations with an uptime of at least 99.9%.

Measurement Criteria: Downtime should be minimal, and any critical failures should be recoverable within 30 minutes.

3.5.3 Availability

Requirement: The system must be available 24/7 with planned maintenance windows.

Measurement Criteria: Maintenance windows should be scheduled during off-peak hours and communicated to users at least 24 hours in advance.

3.5.4 Security

Requirement: The system must ensure secure access, data integrity, and confidentiality.

Measurement Criteria: ○ User authentication should be enforced with strong password policies and two-factor authentication.

Requirement: The system should be easy to maintain and update.

Measurement Criteria:

Code should be well-documented, and the system should follow a modular architecture to facilitate easy updates and bug fixes. Maintenance releases should not require more than 2 hours of downtime.

3.5.5 Portability

Requirement: The system should be portable across different environments (e.g., development, testing, production).

Measurement Criteria: The system should be deployable on various operating systems (e.g., Windows, Linux) without significant changes as it is a web-based system. Use of containerization (e.g., Docker) to ensure consistency across environments.

3.6 Inverse Requirements

Requirement: The system should not allow unauthorized access to sensitive information.

Measurement Criteria: Any attempt to access the system without proper credentials should be logged and trigger an alert to the system administrator.

3.7 Design Constraints

Constraint: The system must comply with organizational policies and procurement regulations.

Measurement Criteria: All procurement workflows must adhere to specified guidelines. The system design should include auditing capabilities to ensure compliance.

3.8 Logical Database Requirements

Requirement: The database must support efficient storage and retrieval of large volumes of data.

Measurement Criteria:

- Database operations (e.g., queries, inserts, updates) should complete within a reasonable timeframe (e.g., less than 1 second for queries).

*Schema***Table: users****Columns:**

<u>id</u>	int AI PK
name	varchar(255)
username	varchar(255)
email	varchar(255)
organization	varchar(255)
location	varchar(255)
reason	text
desired_role	varchar(255)
status	enum('null','accepted','rejected','accepted as admin')
password	varchar(255)

Table: purchaseRequisitions**Columns:**

<u>requisition_id</u>	int AI PK
user_id	int
username	varchar(50)
name	varchar(50)
email	varchar(100)
department	varchar(100)
selected_vendor	varchar(100)
status	enum('null','approved','pending','rejected','closed','complete')
required_on	date
required_by	date
created_at	timestamp

Table: requisition_vendors**Columns:**

<u>requisition_id</u>	int
vendor_name	varchar(100)
vendor_contact_person	varchar(255)
vendor_email_id	varchar(255)
vendor_address	varchar(255)
vendor_GSTIN	varchar(255)
vendor_contact	int
vendor_TIN	varchar(255)
vendor_VAT	varchar(255)
<u>vendor_id</u>	int AI PK

Table: **requisition_items**

Columns:

<u>item_id</u>	int AI PK
requisition_id	int
item_name	varchar(100)
quantity	int
estimated_cost	decimal(12,2)
category_id	int
specification	text
category	varchar(255)

Table: **item_categories**

Columns:

<u>category_id</u>	int AI PK
category_name	varchar(100)
description	varchar(255)

Table: **departments**

Columns:

<u>dept_id</u>	int AI PK
dept_name	varchar(255)
description	varchar(255)

3.9 Other Requirements

Requirement: The system should provide a user-friendly interface that is accessible to users with varying levels of technical expertise.

Measurement Criteria:

- User interfaces should follow usability best practices and be tested with representative users to ensure ease of use.

Chapter 4: *Analysis Models*

4. Analysis Models

4.1 Sequence Diagram

Sequence Diagram: Fetching and Displaying Data in Real-Time
Follow up with **Figure 4.1.1**

Participants:

User: Initiates the creation of purchase requisitions, views approved requisitions, and generates purchase orders.

Administrator: Reviews and approves/rejects purchase requisitions, generates and approves purchase orders.

System: Manages the flow of purchase requisitions and purchase orders, interacts with the database.

Database: Stores and retrieves data related to purchase requisitions and purchase orders.

Sequence of Interactions:

1. *User Creates Purchase Requisition:*

User initiates the creation of a purchase requisition by entering details such as product type, quantity, estimated costs, and potential vendors.

System receives the requisition data and saves it in the **Database**.

System notifies the **Administrator** about the new requisition.

2. *Administrator Reviews and Approves Requisition:*

Administrator reviews the details of the requisition presented by the **System**.

Administrator decides to approve the requisition.

System updates the status of the requisition in the **Database** and notifies the **User** of the approval.

3. *User Views Approved Requisition:*

User requests to view approved requisitions.

System retrieves the requisition data from the **Database**.

System displays the approved requisition details to the **User**.

4. *Admin Generates Purchase Order:*

User initiates the generation of a purchase order based on an approved requisition.

System collects necessary data and generates a purchase order document.

System saves the purchase order in the **Database**.

System notifies the **Administrator** about the newly generated purchase order.

5. *Administrator Reviews and Approves Purchase Order:*

Administrator reviews the details of the purchase order presented by the **System**.

Administrator approves the purchase order.

System updates the status of the purchase order in the **Database** and notifies the **User** of the approval.

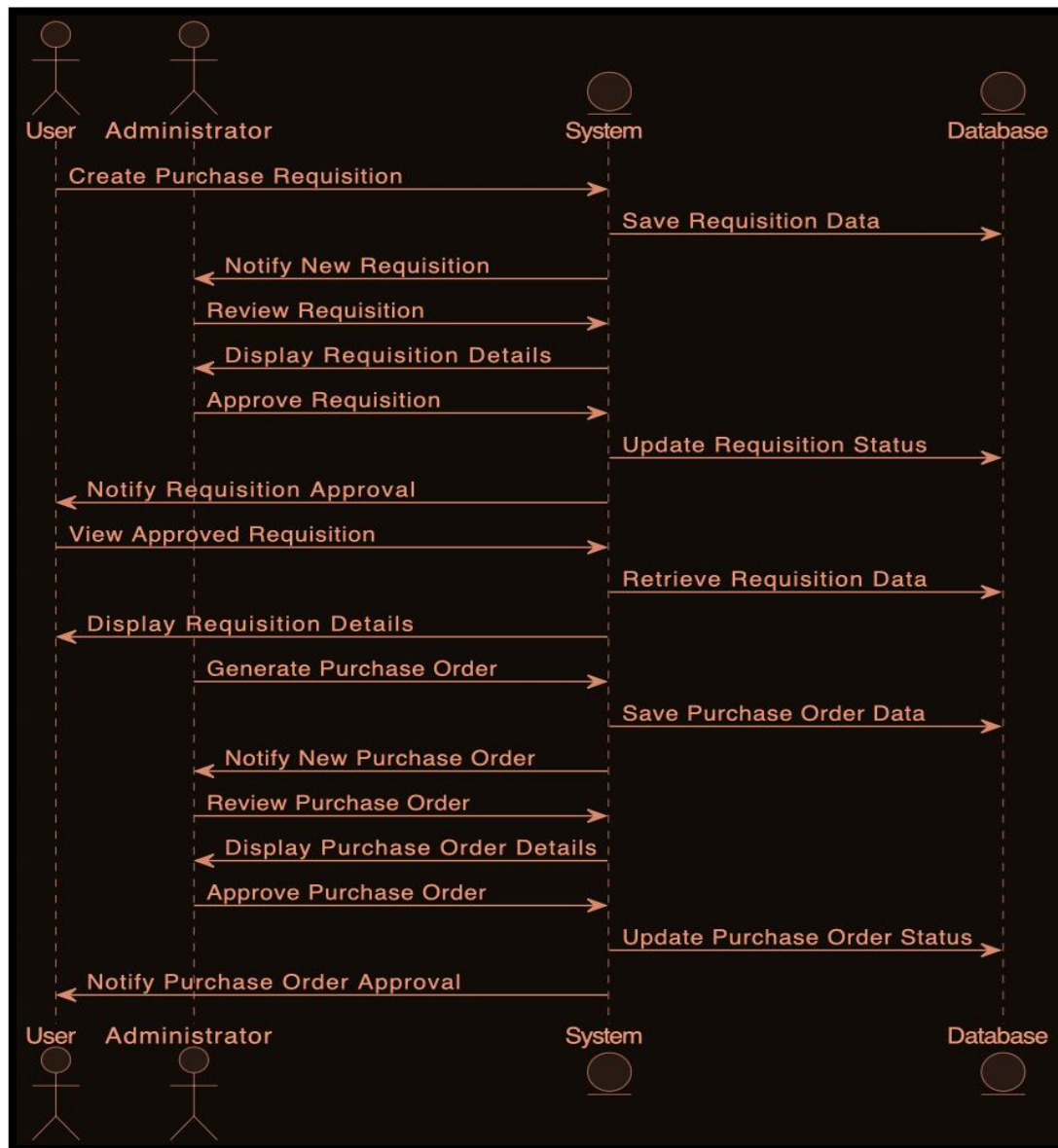


Figure 4.1.1

The workflow begins with the User creating a purchase requisition, which is then reviewed and approved by the Administrator. Once approved, the User can view the requisition and proceed to generate a purchase order. The System ensures all data is properly managed and stored in the Database, with timely notifications sent to relevant parties. Finally, the Administrator reviews and approves the purchase order, completing the process.

4.2 ER Diagram

Entity-Relationship Diagram

Follow up with **Figure 4.2.1**

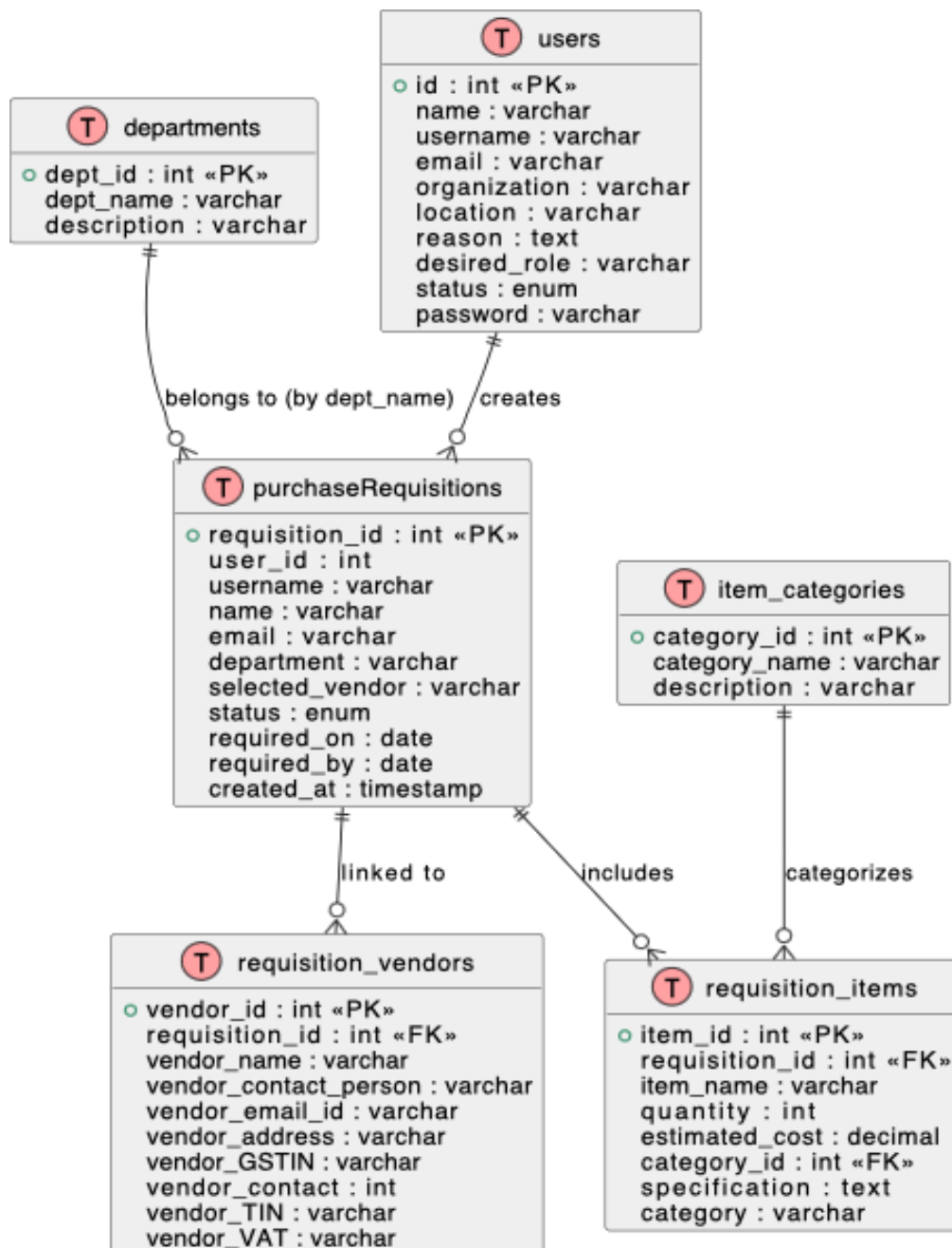
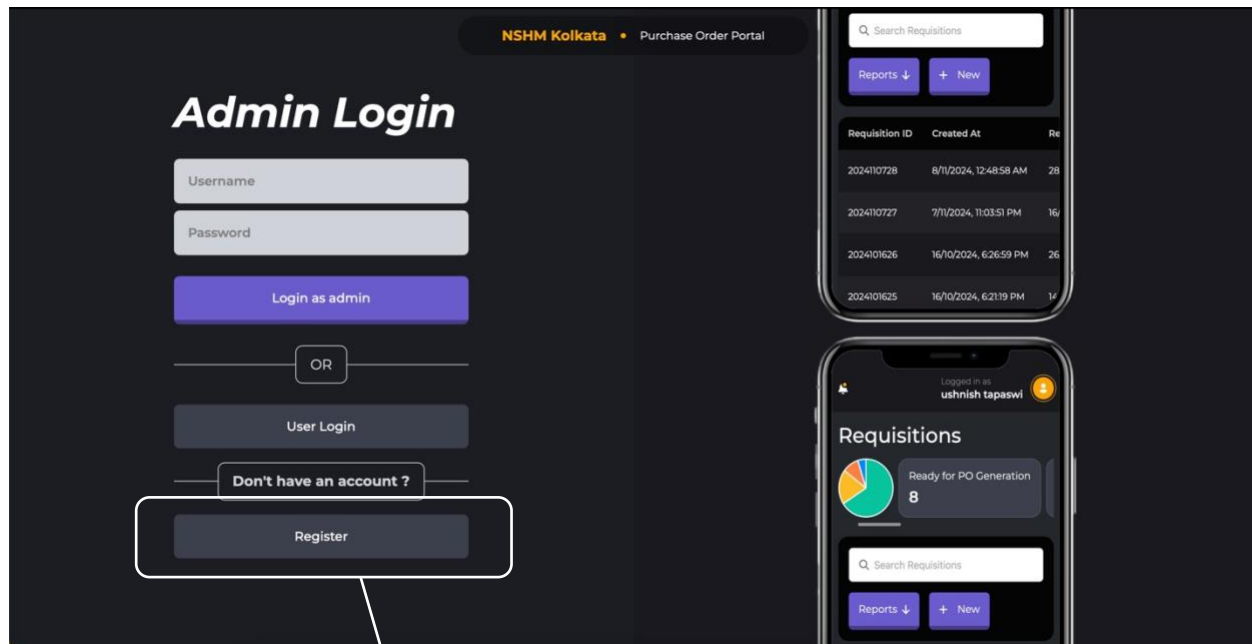


Figure 4.2.1

Chapter 5: *Preview*

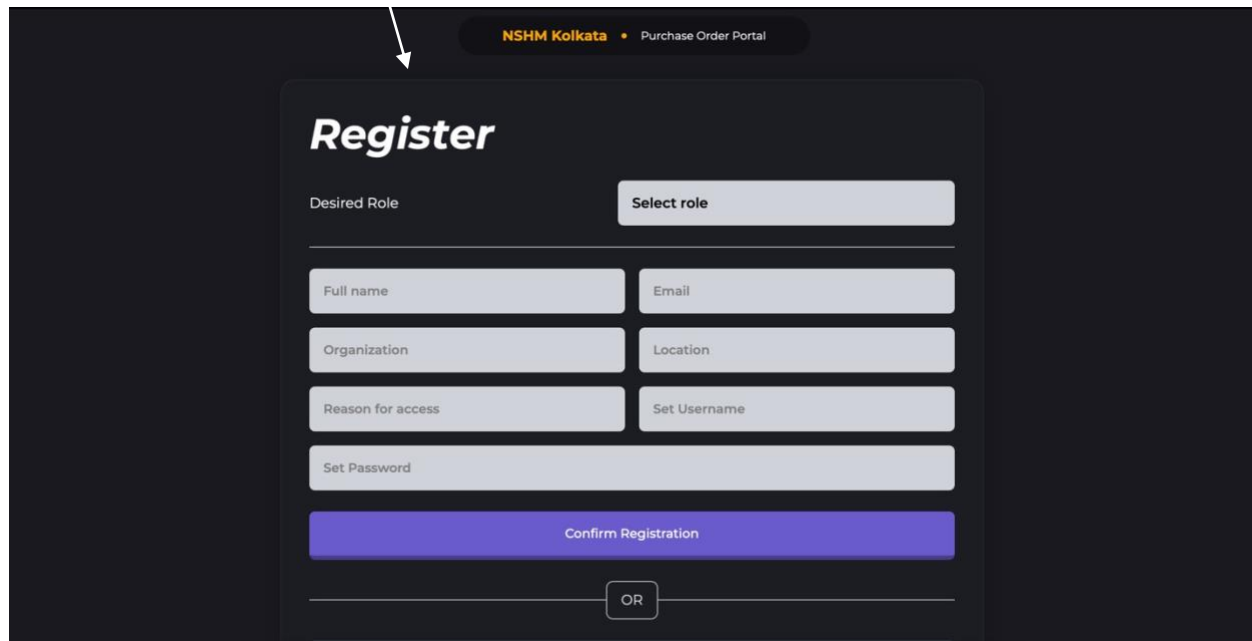
5. Preview

Login & Register



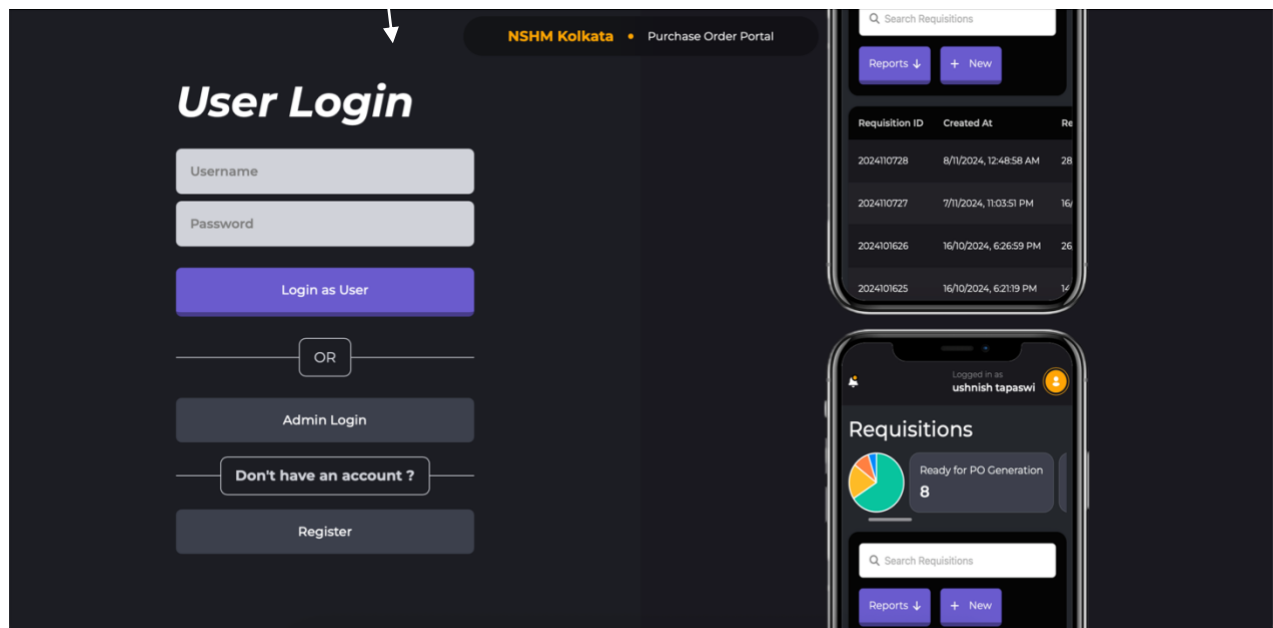
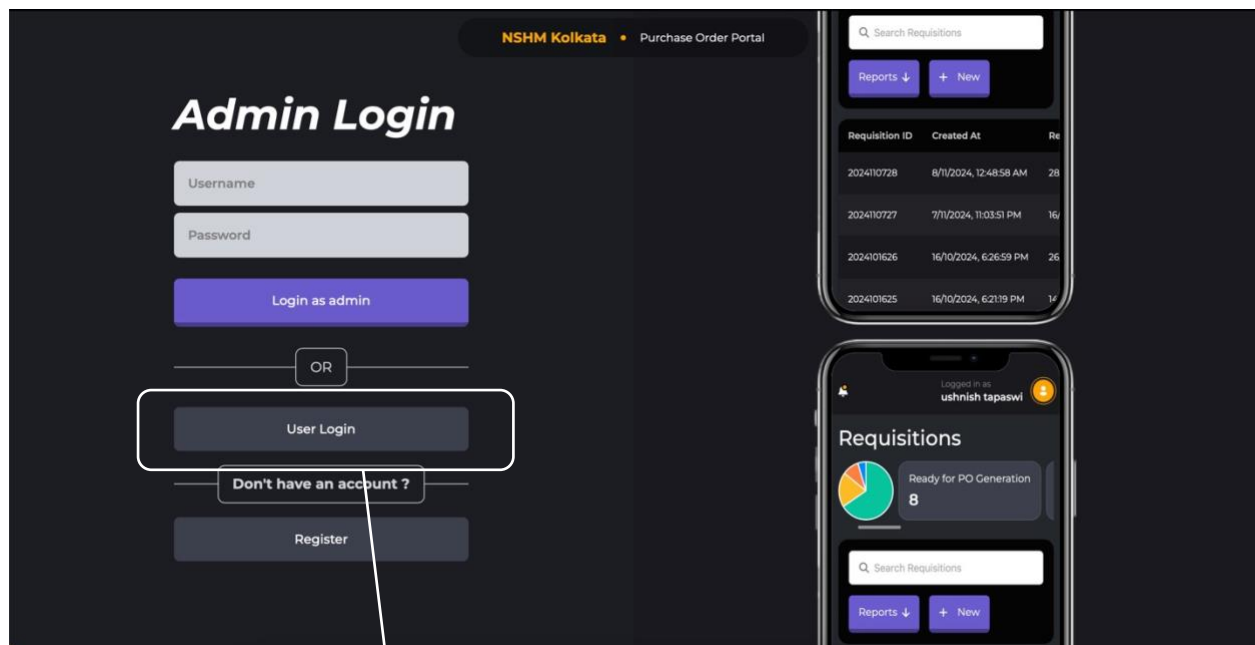
The image shows the 'Admin Login' interface and a mobile app preview. The 'Admin Login' section has a dark background with the title 'Admin Login' in white. It includes input fields for 'Username' and 'Password', a 'Login as admin' button, an 'OR' separator, a 'User Login' button, a 'Don't have an account?' link, and a 'Register' button. The mobile app preview on the right shows a 'Requisitions' screen with a search bar, 'Reports' and 'New' buttons, and a table of requisitions.

Requisition ID	Created At	Re
202410728	8/11/2024, 12:48:58 AM	28
202410727	7/11/2024, 11:03:51 PM	16
2024101626	16/10/2024, 6:26:59 PM	26
2024101625	16/10/2024, 6:21:19 PM	14

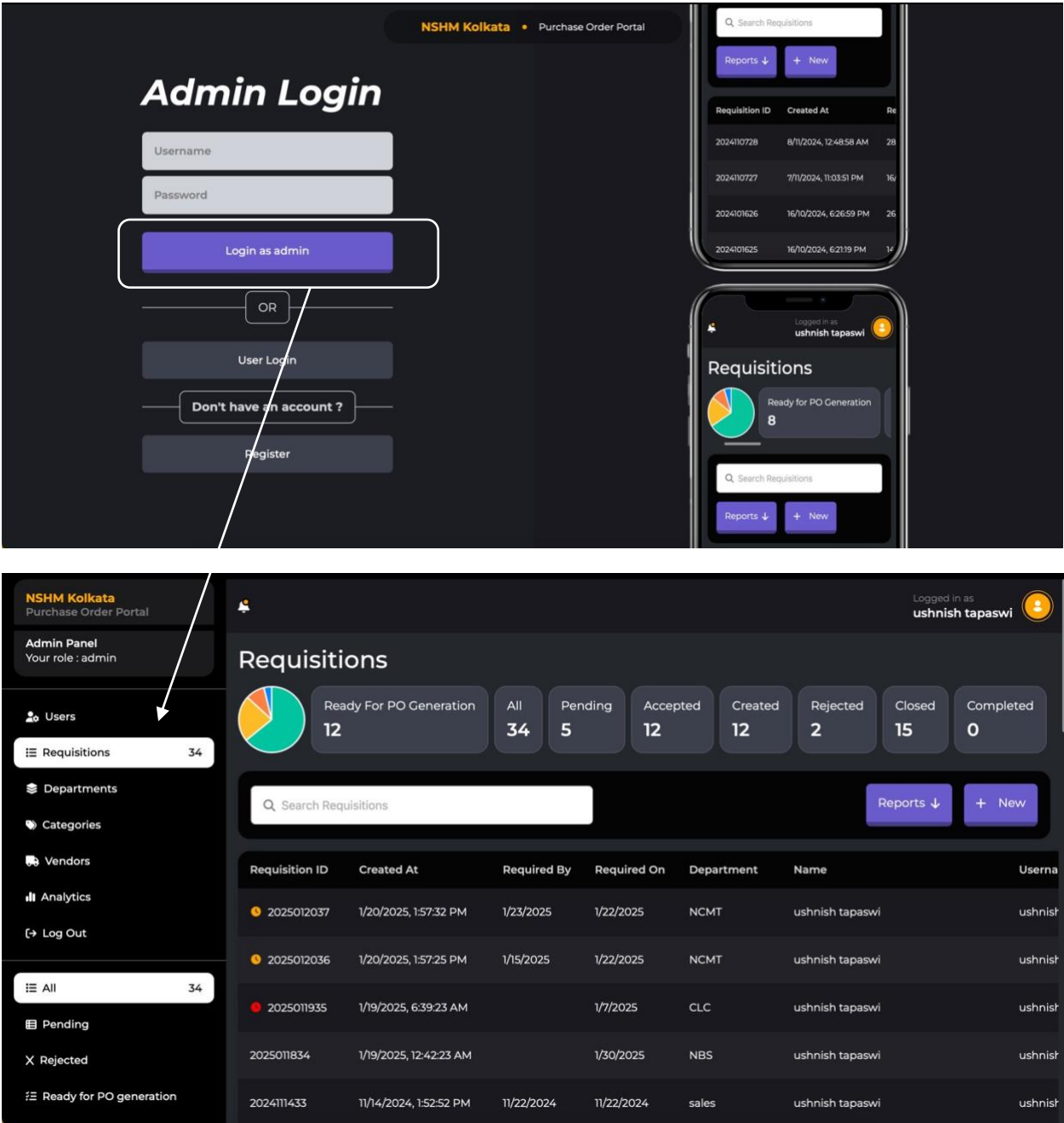


The image shows the 'Register' form with a dark background and the title 'Register' in white. It includes a 'Desired Role' label and a 'Select role' dropdown. Below are input fields for 'Full name', 'Email', 'Organization', 'Location', 'Reason for access', 'Set Username', and 'Set Password'. A 'Confirm Registration' button is at the bottom, followed by an 'OR' separator.

Login & Register



Admin Dashboard - Tables



Admin Dashboard - Tables

NSHM Kolkata

Purchase Order Portal

Admin Panel

Your role : admin

Users

Requisitions 34

Departments

Categories

Vendors

Analytics

Log Out

All 34

Pending

Rejected

Ready for PO generation

Requisitions

Ready For PO Generation 12

All 34

Pending 5

Accepted 12

Created 12

Rejected 2

Closed 15

Completed 0

Search Requisitions

Reports ↓

+ New

Requisition ID	Created At	Required By	Required On	Department	Name	Username
2025012037	1/20/2025, 1:57:32 PM	1/23/2025	1/22/2025	NCMT	ushnish tapaswi	ushnish
2025012036	1/20/2025, 1:57:25 PM	1/15/2025	1/22/2025	NCMT	ushnish tapaswi	ushnish
2025011935	1/19/2025, 6:39:23 AM		1/7/2025	CLC	ushnish tapaswi	ushnish
2025011834	1/19/2025, 12:42:23 AM		1/30/2025	NBS	ushnish tapaswi	ushnish
2024111433	11/14/2024, 1:52:52 PM	11/22/2024	11/22/2024	sales	ushnish tapaswi	ushnish

NSHM Kolkata

Purchase Order Portal

Admin Panel

Your role : admin

Users 4

Requisitions

Departments

Categories

Vendors

Analytics

Log Out

Active Users

Pending Requests

Users

Active Users 4

Pending 4

Admins 3

General Users 1

Search User

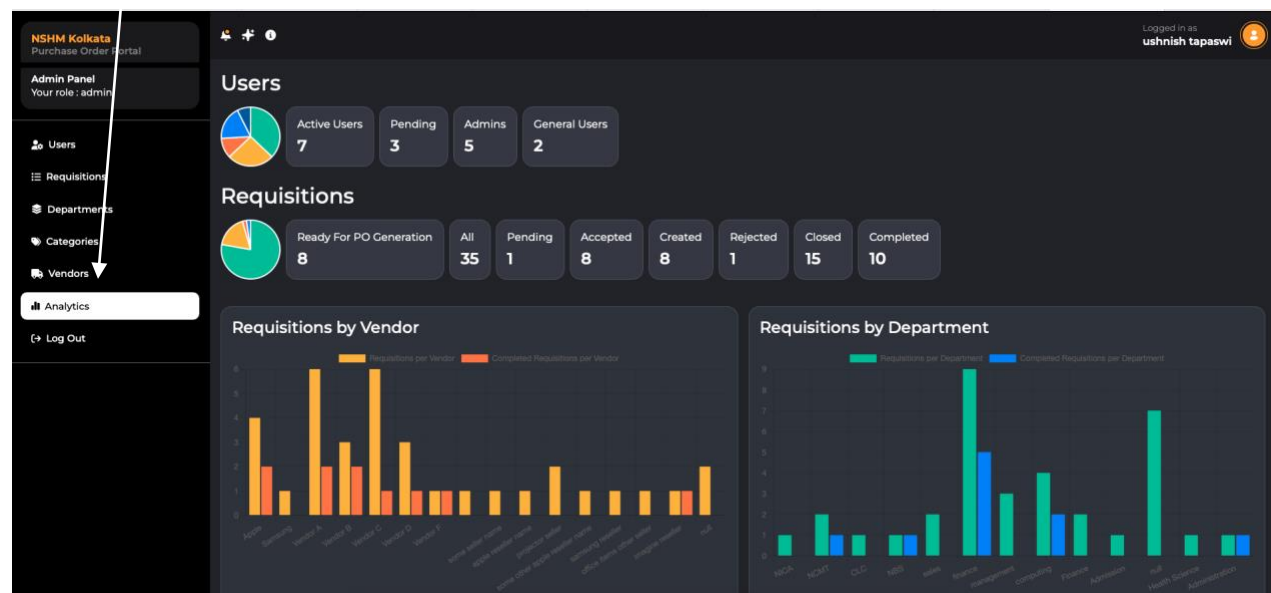
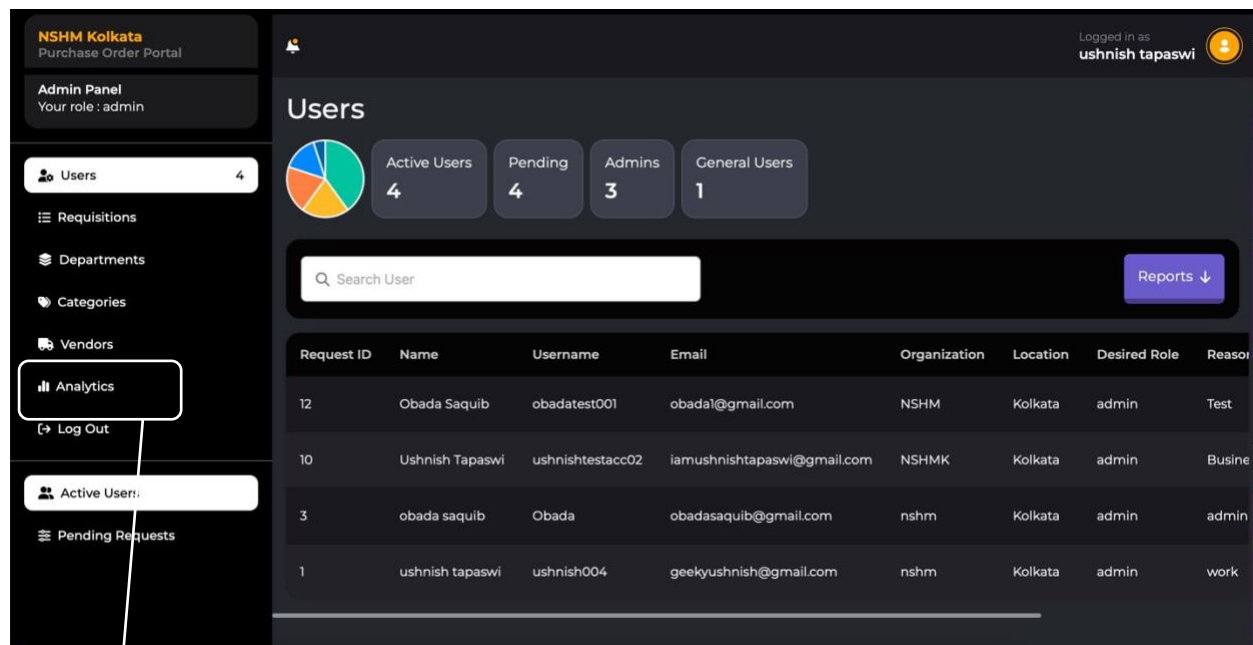
Reports ↓

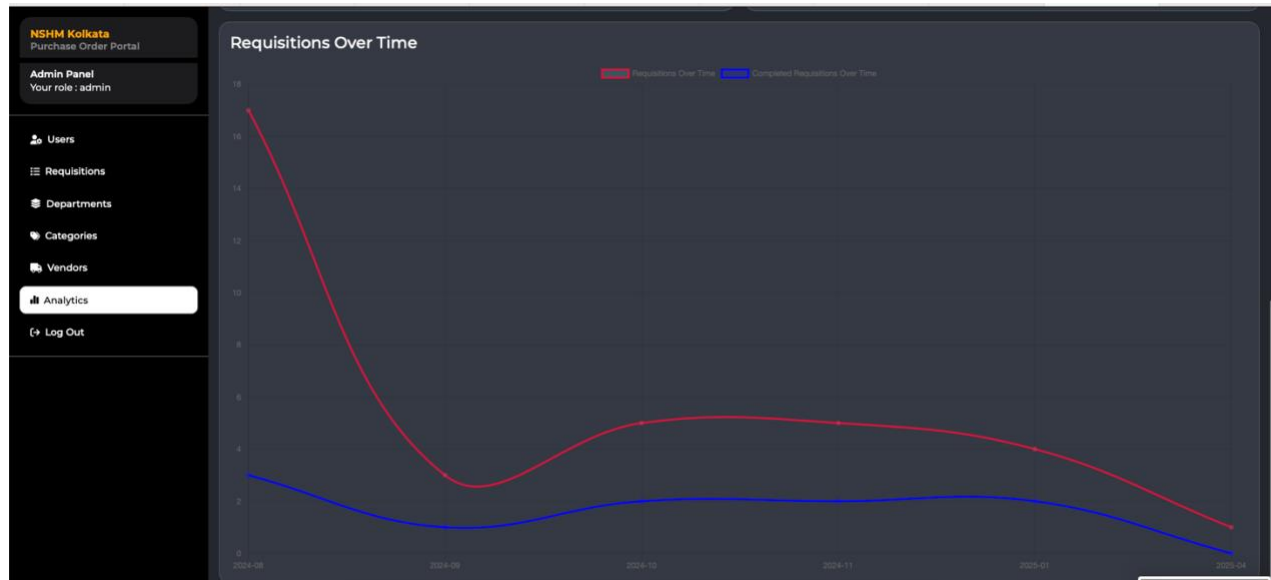
Request ID	Name	Username	Email	Organization	Location	Desired Role	Reason
12	Obada Saquib	obadatest001	obada1@gmail.com	NSHM	Kolkata	admin	Test
10	Ushnish Tapaswi	ushnishtestacc02	iamushnishtapaswi@gmail.com	NSHMK	Kolkata	admin	Business
3	obada saquib	Obada	obadasaquib@gmail.com	nshrm	Kolkata	admin	admin
1	ushnish tapaswi	ushnish004	geekyushnish@gmail.com	nshrm	Kolkata	admin	work

43

Purchase Order System

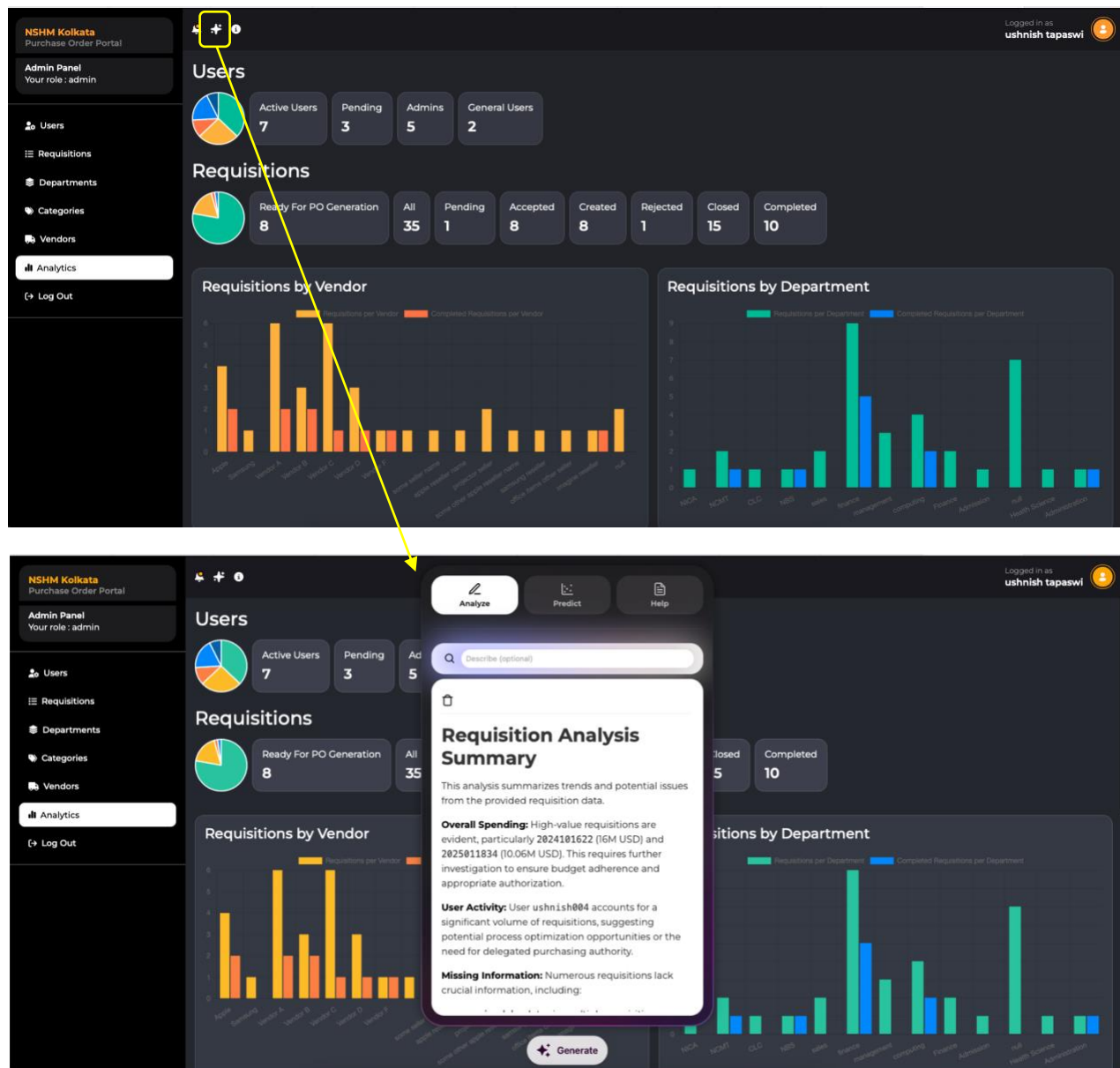
Admin Dashboard – Analytics





Purchase Order System

Admin Dashboard – Integrated LLM helper



Modals – Notification Centre, Add/Edit Requisitions

NSHM Kolkata
Purchase Order Portal

Admin Panel
Your role : admin

Users

Requisitions 34

Departments

Categories

Vendors

Analytics

Log Out

All 34

Pending

Rejected

Ready for PO generation

Requisitions

Ready For PO Generation 12

All 34

Pending 5

Accepted 12

Created 12

Rejected 2

Closed 15

Completed 0

Search Requisitions

Reports ↓

+ New

Requisition ID	Created At	Required By	Required On	Department	Name	Username
2025012037	1/20/2025, 1:57:32 PM	1/23/2025	1/22/2025	NCMT	ushnish tapaswi	ushnist
2025012036	1/20/2025, 1:57:25 PM	1/15/2025	1/22/2025	NCMT	ushnish tapaswi	ushnist
2025011935	1/19/2025, 6:39:23 AM		1/7/2025	CLC	ushnish tapaswi	ushnist
2025011834	1/19/2025, 12:42:23 AM		1/30/2025	NBS	ushnish tapaswi	ushnist
2024111433	11/14/2024, 1:52:52 PM	11/22/2024	11/22/2024	sales	ushnish tapaswi	ushnist

Notifications Done

LATEST

ushnish004 created a new requisition (id) 2025012037
1/20/2025, 1:57:32 PM

1 requisitions require action

2 requisitions require immediate action

15 requisitions are past deadline

Add New Requisition

Item Name

Item Category

Select Category

Specifications

Quantity

Estimated Cost

Add Item +

Department:

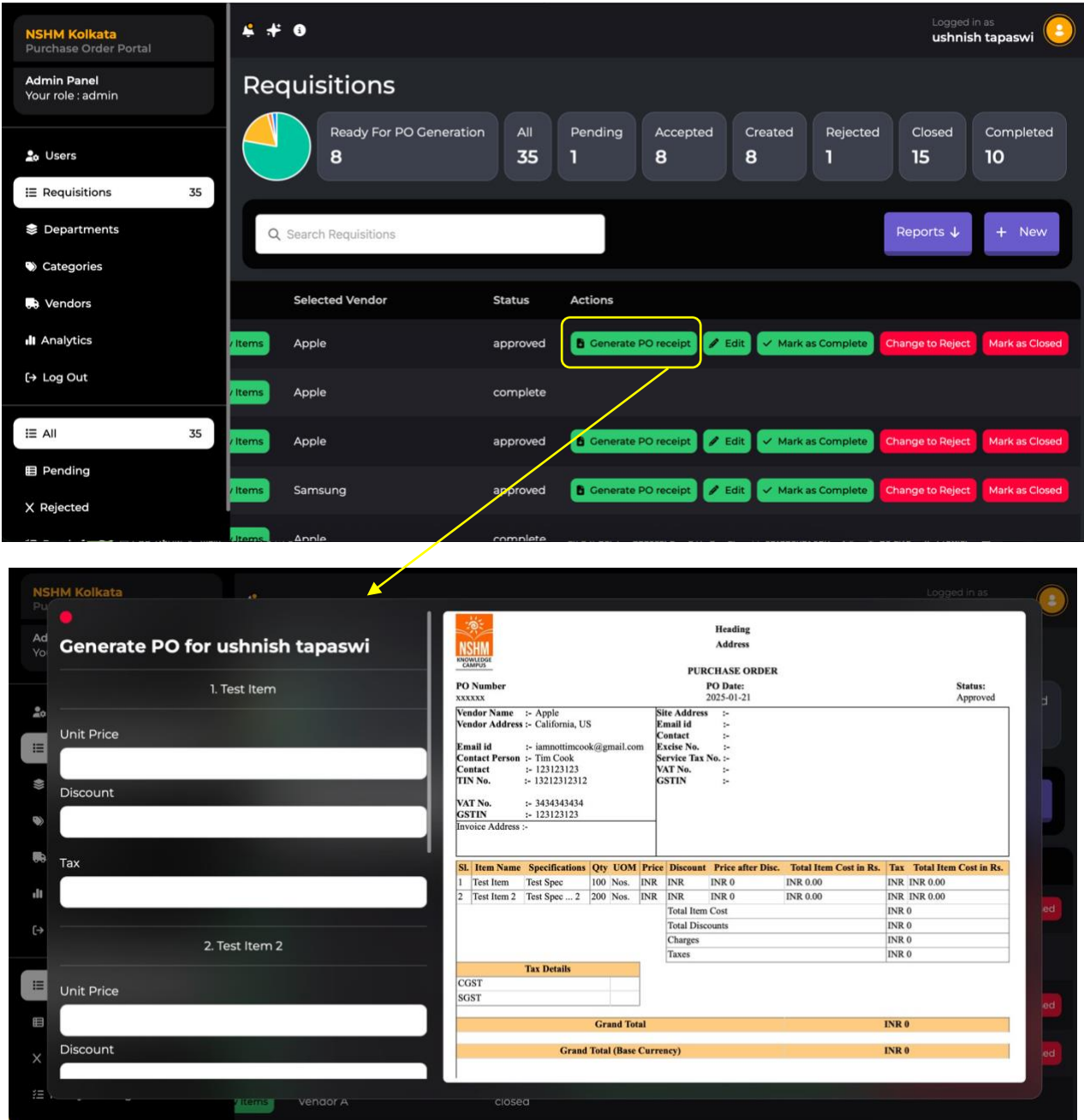
Select Department

Preferred Vendor:

Select Vendor

Required On:

Admin Dashboard – PO generation modal



Appendix

Appendix A: Abbreviations and Acronyms

POS: Purchase Order System

SRS: Software Requirements Specification

RFQ: Request for Quotes

PO: Purchase Order

NSHM: Nabadigant School of Higher Management

UI: User Interface

API: Application Programming Interface

HTTP: Hypertext Transfer Protocol

HTTPS: Hypertext Transfer Protocol Secure

Appendix B: Glossary

Purchase Requisition	A formal internal request made by an employee or department to procure goods or services. It initiates the procurement process.
Vendor	A company or individual that supplies goods or services to an organization. In this system, vendors can be added, viewed, or deactivated.
Vendor Management	The process of managing supplier details, qualifications, and relationships to ensure reliable supply of goods/services.
Request for Quote (RFQ)	A document or email sent to vendors to request price quotations for specified items or services.
Purchase Order (PO)	A commercial document issued by the buyer to the vendor indicating types, quantities, and agreed prices for products or services.
Approval Workflow	A predefined process through which a purchase request or order moves for necessary authorization by one or more approvers.

Dashboard	A user interface that displays key performance indicators, system activity, and user-specific functions in a consolidated view.
Admin Portal	A secure interface used by administrators to manage users, vendors, requisitions, approvals, and reports.
General User	A user role responsible for submitting requisitions and tracking their status but with no administrative privileges.
System Administrator	A user with elevated permissions who can manage all aspects of the system, including user roles, approvals, and configuration.
Authentication	The process of verifying a user's identity, typically using credentials such as a username and password.
Authorization	The process of determining which actions a user is permitted to perform within the system after successful authentication.
Notification	A system-generated alert (either on-screen or via email) to inform users about actions or status changes such as approvals or rejections.
Analytics	The interpretation of data generated by the system, often displayed through dashboards to help users make informed decisions.
Role-Based Access Control (RBAC)	A security principle that restricts system access based on a user's role, ensuring least-privilege access.
Sequence Diagram	A type of UML diagram showing how processes operate with one another and in what order.
ER Diagram	Entity-Relationship Diagram, a visual representation of data objects and relationships between them in a database system.
Frontend	The client-side part of the application that users interact with, usually built using HTML, CSS, JavaScript, or React.
Backend	The server-side of the application responsible for handling logic, database operations, and API communication.
Database	A structured set of data held in a computer, especially one that is accessible in various ways (e.g., MySQL, PostgreSQL).
UI (User Interface)	The layout and interactive elements through which users interact with the software.
API (Application Programming Interface)	A set of rules and tools for building software applications, enabling different components to communicate.
HTTPS	A secure version of HTTP used to encrypt data exchanged between the user and server.
Two-Factor Authentication (2FA)	An added security measure that requires users to verify their identity with a second factor (e.g., OTP).
Export (Reports/PO)	The action of downloading system-generated reports or documents in formats like PDF or Excel.

Appendix C: Sample Data

Sample User Accounts:

Admin: admin@example.com | Password: Admin@123

General User: user@example.com | Password: User@123

Sample Vendors:

Vendor Name: ABC Supplies, Contact: vendor@abc.com

Vendor Name: XYZ Traders, Contact: contact@xyz.com

Sample Purchase Requisitions:

Requisition ID	User	Item	Qty	Estimated Cost	Potential Vendor	Status
PR001	user1	Dell Laptops	5	₹3,00,000	V001, V003	Approved
PR002	user1	Office Chairs	10	₹50,000	V002	Pending
PR003	manager1	Whiteboards	3	₹12,000	V002	Rejected

Purchase Order System

Sample Purchase Order Receipt :

NSHM KNOWLEDGE CAMPUS		Sample Heading Sample Address		PURCHASE ORDER		PO Number Sample PO Number		PO Date: 2025-06-16		Status: Approved	
Vendor Name :- Vendor B		Site Address :- sample address		Email id :- sample@sample.com		Contact :- 1283709189		Excise No. :- 817112311		Service Tax No. :- 12381918991	
Vendor Address :-		Email id :-		Contact Person :-		Contact :-		VAT No. :- 123118371879		GSTIN :- 12381250986	
Email id :-		Invoice Address :-									
VAT No. :-											
GSTIN :-											
SL	Item Name	Specifications	Qty	UOM	Price	Discount	Price after Disc.	Total Item Cost in Rs.	Tax	Total Item Cost in Rs.	
1	iPhone	256gb purple	30	Nos.	INR 70000	INR 3000	INR 67000	INR 2010000.00	INR 989.87	INR 2010989.87	
2	iMac	1TB Ram 2TB ROM	30	Nos.	INR 77000	INR 5000	INR 72000	INR 2160000.00	INR 568.56	INR 2160568.56	
3	iPad	M4	50	Nos.	INR 35000	INR 2000	INR 33000	INR 1650000.00	INR 329.28	INR 1650329.28	
Total Item Cost								INR 5823186.71			
Total Discounts								INR 340000			
Charges								INR 1299			
Taxes								INR 1887.7099999999998			
Tax Details											
CGST			0								
SGST			0								
Grand Total : Rupees Fifty Eight Lakh Twenty Three Thousand One Hundred and Eighty Six and and Seventy One Paise only								INR 5823186.71			
Grand Total : (Base Currency) Rupees Fifty Eight Lakh Twenty Three Thousand One Hundred and Eighty Seven only								INR 5823187			
Terms & Conditions											
1. Term 1											
1. Term 2											
Purchase Clauses											
1. Clause 1											
1. Clause 2											
Created by Ushnish Tapaswi						Acknowledged by Ushnish Tapaswi					

Appendix D: System Configuration

Minimum System Requirements:

Browser: Latest Chrome, Firefox, Safari

RAM: 2GB minimum

Network: Stable internet connection with a speed of 1 Mbps or higher

Server Requirements:

CPU: Quad-core processor

RAM: 2GB minimum

Database: MySQL/PostgreSQL

Hosting: Linux-based server with Apache/Nginx

Appendix E: References

1. IEEE Standard 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
2. ISO/IEC 25010:2011 – Systems and Software Engineering — System and Software Quality Requirements and Evaluation (SQuaRE)

3. OWASP Secure Coding Practices (<https://owasp.org/www-project-secure-coding-practices/>)
4. Nielsen Norman Group – UI/UX Design Principles
(<https://www.nngroup.com/articles/ten-usability-heuristics/>)
5. REST API Design Guidelines – Microsoft (<https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design>)
6. React Documentation (<https://reactjs.org/>)
7. SQL Documentation (<https://www.sql.org/>)
8. Node.js Documentation (<https://nodejs.org/en/docs/>)
9. Express.js Documentation (<https://expressjs.com/>)
10. Gemini Documentation (<https://ai.google.dev/gemini-api/docs>)

Appendix F: Statement of Originality

We, the undersigned, hereby declare that this project report titled "Purchase Order System – Web Application with Admin Portal" is our original work and has not been submitted previously for any academic award. All sources and references have been appropriately acknowledged.

End of doc.